

SupplyChain Nexus Modules

1. Data Upload & Integration

- Upload all Excel datasets
- Merge datasets automatically
- Validate data
- Detect missing values, duplicates and inconsistencies
- Generate Data Quality Score

2. Dashboard

- Total Warehouses
- Total Customers
- Total Demand
- Total Capacity
- Warehouse Health Score
- Optimization Score
- Cost
- Distance
- Travel Time
- Risk Index

3. Warehouse Intelligence

- Capacity Utilization
- Available Capacity
- Assigned Demand
- Warehouse Efficiency
- Warehouse Health Score
- Warehouse Ranking

4. Customer & Demand Intelligence

- Customer Distribution
- Demand Analysis
- K-Means Clustering
- Demand Hotspots
- Warehouse Influence Zones

5. GIS Map

- Warehouse Locations
- Customer Locations
- Optimized Routes
- Heatmaps
- Cluster Visualization
- Route Visualization
- Risk Visualization

6. Multi-Objective Optimization Engine (Core)

Using Google OR-Tools

Optimize for:

- Minimum Transportation Cost
- Minimum Travel Distance
- Minimum Travel Time
- Balanced Warehouse Utilization
- Capacity Constraints
- Cold Storage Constraints

7. Optimization Profiles

- Cost Saver
- Fast Delivery
- Balanced
- Sustainability
- Emergency Mode
- Custom Weights

8. Scenario Analysis

Simulate:

- Warehouse Failure
- Capacity Reduction
- Demand Increase
- Demand Decrease
- Fuel Cost Increase
- Route Closure
- Vehicle Shortage
- Cold Storage Failure

Compare Before vs After.

9. AI Recommendation Engine

Generate recommendations like:

- Shift customers
- Change warehouse
- Reduce overload
- Improve utilization
- Reduce transportation cost
- Reduce travel time

Each recommendation includes:

- Reason
- Confidence
- Expected Savings
- Business Impact

10. Explainable AI

Explain:

- Why warehouse changed
- Why recommendation generated
- Expected improvement
- Optimization logic

11. Risk Detection

Isolation Forest

Detect:

- Overloaded Warehouses
- Cost Outliers
- Route Anomalies
- Capacity Issues
- Demand Anomalies

12. Warehouse Health Report

For every warehouse:

- Health Score
- Utilization
- Risk
- Capacity
- AI Suggestions

- Performance Summary

13. Executive Decision Center

Business KPIs:

- Cost
- Distance
- Time
- Utilization
- Savings
- Carbon Emissions
- Optimization Score
- Supply Chain Health

14. AI Insights

Automatically generate:

- Optimization Summary
- Warehouse Warnings
- Bottlenecks
- Cost Saving Opportunities
- Underutilized Warehouses

15. Digital Twin

Animated logistics simulation showing:

- Warehouse Operations
- Route Flow
- Customer Allocation
- Optimization Replay
- Scenario Simulation

16. Optimization Replay

Show:

- Initial Allocation
- Optimization Steps
- Final Allocation

Animated replay with timeline.

17. Analytics

Charts:

- Demand Distribution
- Warehouse Utilization
- Cost Analysis
- Distance Analysis
- Travel Time
- Risk Distribution
- Scenario Comparison

18. Sustainability

Calculate:

- Fuel Consumption
- CO₂ Emissions
- Green Logistics Score
- Distance Saved
- Carbon Reduction

19. AI Decision Copilot

Ask questions like:

- Which warehouse is overloaded?
- Why did optimization choose Warehouse W3?
- Which scenario is best?
- How much cost was saved?

20. Smart Constraint Builder

Users can enable or disable:

- Capacity Constraints
- Budget Constraints
- Travel Time Limits
- Cold Storage Limits
- Distance Limits

21. Notifications

Alerts:

- High Utilization
- Route Issues
- Optimization Complete
- Data Quality Issues
- High Risk Warehouses

22. Reports

Generate:

- Executive Report
- Optimization Report
- Warehouse Report
- Risk Report
- Scenario Report

- Sustainability Report

Export:

- PDF
- Excel
- CSV

23. Mission Control Dashboard

A full-screen live dashboard showing:

- KPIs
- Maps
- Routes
- Digital Twin
- Alerts
- Optimization Status

Unique Features (USP)

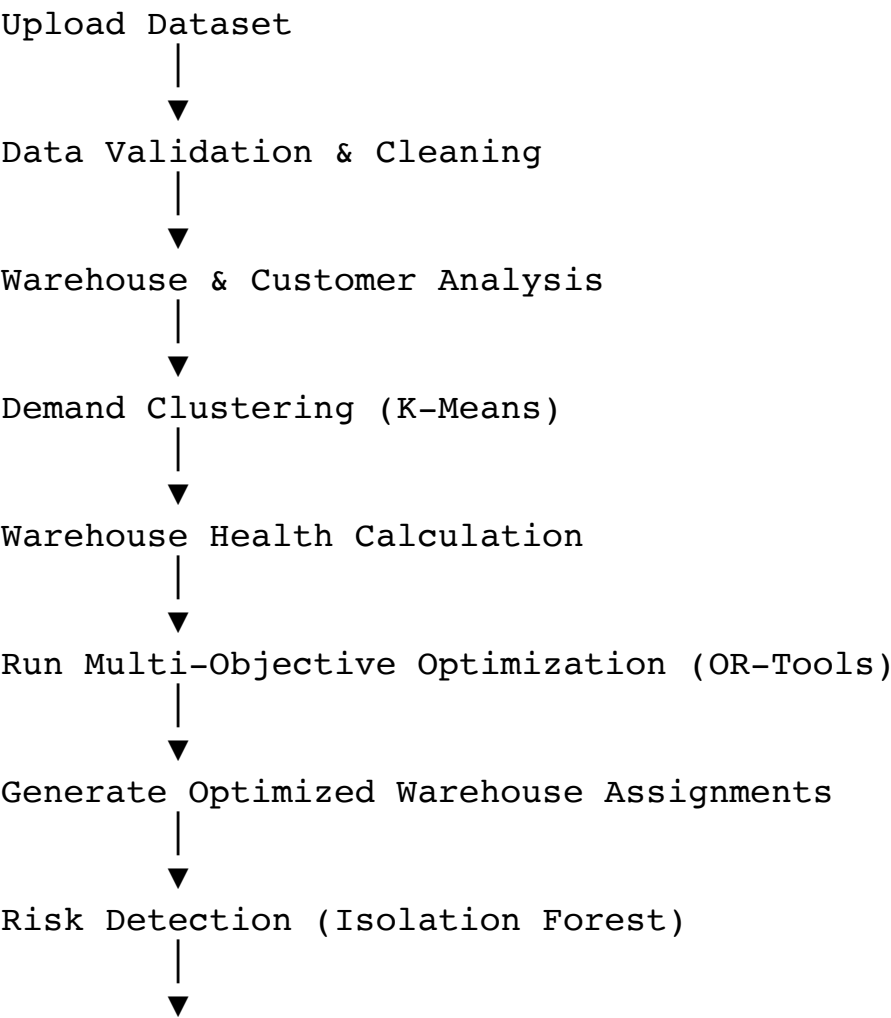
- Warehouse Health Score
- Optimization Replay
- Digital Twin Visualization
- AI Decision Copilot
- Executive Decision Center
- Scenario Laboratory
- Before vs After Comparison
- Green Logistics Dashboard
- Benchmark Different Optimization Strategies
- Explainable AI Recommendations
- Risk Propagation Visualization
- Smart Constraint Builder

- Interactive KPI Wall
- Professional Enterprise Dashboard

AI / Optimization Models Used

Module	Model
Customer Clustering	K-Means
Risk Detection	Isolation Forest
Route Optimization	Google OR-Tools
Network Analysis	NetworkX
GIS Processing	GeoPandas
Recommendation Engine	Rule-based + Optimization Results
Scenario Analysis	OR-Tools Re-optimization

Overall Workflow



AI Recommendations & Explainable Insights



Scenario Simulation & Re-optimization



Interactive Maps & Digital Twin



Executive Dashboard & Reports

Final Goal

SupplyChain Nexus acts as an **AI-assisted supply chain decision support platform**. It transforms raw logistics data into optimized warehouse allocations, scenario-based planning, explainable recommendations, and interactive visualizations, enabling organizations to make faster, data-driven decisions that reduce costs, improve efficiency, and increase supply chain resilience.